

PardonMix: Unpacking Ambiguity in Semantic Audio Mixing for Adaptive Interface Design

Dongwoo Kim

Human-Centered Intelligent Systems Lab
Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
dwjustin@gm.gist.ac.kr

Jeongseok Oh

Human-Centered Intelligent Systems Lab
Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
jungseok17@gm.gist.ac.kr

Kangbeen Ko

Human-Centered Intelligent Systems Lab
Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
kangbeen.ko@gm.gist.ac.kr

SeungJun Kim*

Human-Centered Intelligent Systems Lab
Gwangju Institute of Science and Technology
Gwangju, Republic of Korea
seungjun@gist.ac.kr

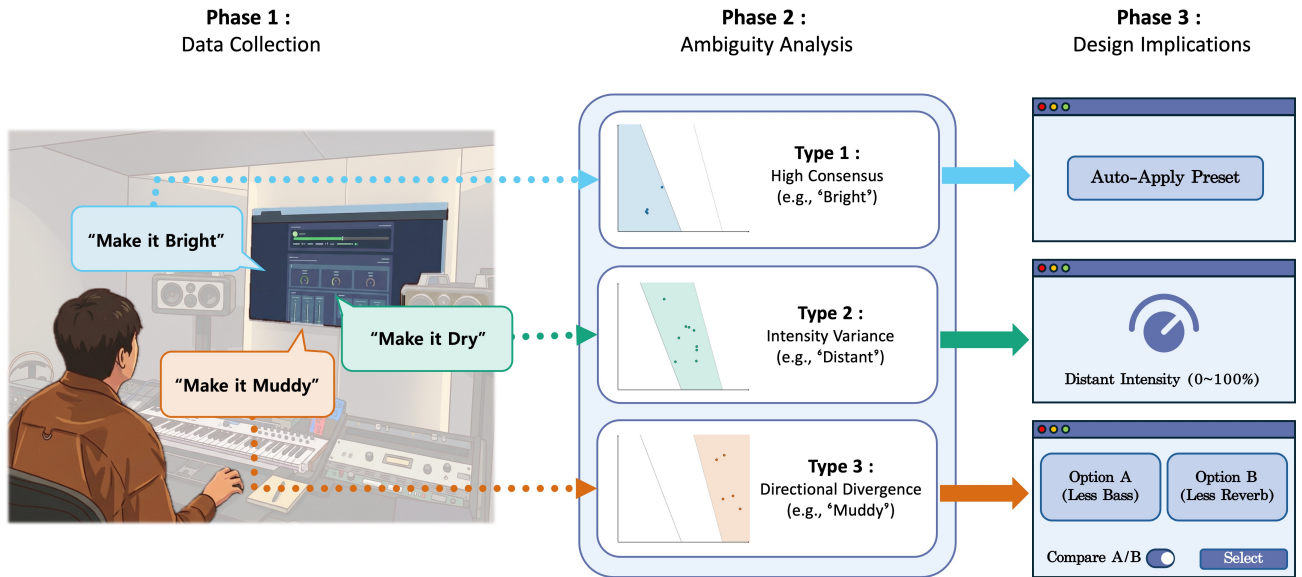


Figure 1: Overview of the proposed Adaptive Semantic Mixing Framework. The study analyzes professionals' *interpretative diversity* to classify abstract descriptors into three ambiguity types—(1) High Consensus, (2) Intensity Variance, and (3) Directional Divergence—and dynamically adapts interface strategies to support expert–novice communication by bridging semantic gaps and aligning creative intent with technical execution.

Abstract

Audio mixing requires sophisticated parameter control and accumulated listening experience. In collaborative workflows, novice musicians often struggle to convey their auditory intent to professional engineers, creating a semantic gap between abstract descriptors

and technical execution. While prior studies have mapped descriptive terms to parameters by focusing on minimizing ambiguity, we propose that identifying types of ambiguity can serve as a key to bridging this communication gap. We conducted a formative study analyzing mixing parameter vectors from 10 professional engineers for 18 abstract descriptors, identifying distinct types of ambiguity: High Consensus, Intensity Variance, and Directional Divergence. We propose PardonMix, an adaptive interface strategy that dynamically allocates disambiguation widgets; direct automation for high consensus terms, degree control for intensity varying terms, and exploratory gallery for divergent terms. PardonMix seeks to bridge the semantic gap by embracing ambiguity, facilitating smoother

*Corresponding Author



This work is licensed under a Creative Commons Attribution 4.0 International License.
CHI EA '26, Barcelona, Spain

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3798647>

collaboration that ensures the novice's artistic vision is precisely translated into technical execution.

CCS Concepts

• **Applied computing** → **Sound and music computing**; • **Human-centered computing** → *Empirical studies in interaction design*.

Keywords

Audio mixing, Semantic audio, Creativity support tools, Adaptive interface, Interpretation Ambiguity

ACM Reference Format:

Dongwoo Kim, Kangbeen Ko, Jeongseok Oh, and SeungJun Kim. 2026. PardonMix: Unpacking Ambiguity in Semantic Audio Mixing for Adaptive Interface Design. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3772363.3798647>

1 Introduction

Audio mixing, a critical stage in music post-production [5], entails adjusting levels and timbre to integrate recorded tracks into a cohesive song [22, 31]. Audio mixing is a complex engineering task far more than simple volume balancing, requiring knowledge and accumulative listening experience [6, 10, 35]. For instance, ensuring a vocal sits clearly atop a beat demands not just leveling, but meticulously carving out frequency ranges where instruments compete for space (i.e., masking), or precisely shaping transients to prevent dynamic clashes between the kick drum and bass [31]. Consequently, novice musicians adopt a remote collaborative workflow by outsourcing to experts, yet unlike skilled practitioners, they struggle to convey auditory intent as they rely solely on abstract natural language [18, 37]. In this process, a *semantic gap* arises between the novice's non-professional language and the expert's technical language [6, 37].

Across domains ranging from robotics [13] to software engineering [36], translating ambiguous natural language into executable commands has been a long-standing challenge. In music production, semantic audio research has similarly sought to map abstract descriptors to technical parameters [6, 35]. However, unlike recent advances in NLI or generative design that operationalize uncertainty through mixed-initiative widgets [15, 26], mixing systems predominantly enforce a deterministic, one-to-one mapping (e.g., the mean) [34]. This approach treats interpretative ambiguity as statistical variance to be minimized, which may not fully capture the reality that a term like *Heavy* can trigger divergent strategies, such as spectral boost versus dynamic compression [35]. Such rigidity limits artistic agency [19], creating a critical gap where mixing interfaces lag behind broader HCI trends that embrace ambiguity as a productive resource for sensemaking [16]. Addressing this gap, we propose *PardonMix*, an adaptive interface strategy that integrates variance into a design perspective. We first unpack the structure of ambiguity by analyzing mixing parameter vectors from 10 professional engineers, identifying three distinct types: *High Consensus*, *Intensity Variance*, and *Directional Divergence*. Leveraging these insights, we introduce an adaptive strategy that dynamically deploys disambiguation widgets—ranging from automation to exploratory galleries—contingent upon the detected ambiguity type.

2 Methodology

2.1 Participants

We recruited 10 professional mixing engineers with at least 3 years of paid experience (0 female, 10 male; $M_{age} = 31.1$ years, $SD_{age} = 6.21$; $M_{paid} = 8.1$ years, $SD_{paid} = 5.17$), a minimum of 10 hours of weekly mixing work, and a portfolio of over 20 commercial projects each. Participants were compensated \$25 for their time and contribution to the dataset construction. This study was approved by the Institutional Review Board (IRB).

2.2 Experimental Setup

We built a custom web-based mixing interface using the Web Audio API, ensuring sample-accurate timing for perceptual evaluation [23]. This allowed us to observe how experts translate abstract descriptors into technical parameter adjustments, while minimizing confounds introduced by differences across commercial mixing tools.

As illustrated in Fig. 2, the interface provided controls across three dimensions: Spectral, Dynamics, and Spatial [31]. The Spectral module featured a 3-band equalizer (Low/Mid/High) with a range of ± 12 dB. The Dynamics module included a compressor with adjustable Threshold (-30 to 0 dB), Attack (0 to 200 ms), and Gain to manage loudness density. The Spatial module allowed manipulation of Reverb (Mix/Decay) and Delay (Mix) to shape the acoustic environment. To lower the barrier, the UI adopts skeuomorphic metaphors (rotary knobs, sliders) that mimicking physical audio hardware familiar to musicians [3]. The interface incorporated an **Audio Player** (Fig. 2-A) which loops the stimulus and displays the target descriptive term. Additionally, the **Utility tools** (Fig. 2-B) provide a **Bypass** option, allowing users to instantly toggle between the processed signal and the raw audio for A/B comparison, and a **Reset** function to restore all parameters to their neutral default states (e.g., 0 dB gain, 0% reverb mix) for rapid error recovery during trials.

To ensure a neutral starting point for mixing, we utilized a 16-second multitrack loop (117 BPM) comprising four dry stems: drum, bass, guitar riff, and vocal. High-resolution raw samples were selected from the Splice library, prioritizing acoustic realism over synthesized sounds [12]. Prior to the task, all tracks were normalized to -23 LUFS (integrated) following ITU-R BS.1770-4 standards [21] to guarantee consistent loudness perception.

2.3 Vocabulary Selection

To define the mixing goals for the participants, we curated a list of 18 descriptive adjectives strictly derived from established literature based on two criteria: (1) whether the term is frequently used in real-world mixing requests [42], and (2) whether the term has the potential for interpretative divergence among engineers [6, 35, 42], making it suitable for observing ambiguity in mixing contexts. Accordingly, the selected terms were categorized into three dimensions corresponding to our interface's controls: **Spectral (EQ)**, **Spatial**, and **Dynamic**. The final set consisted of 10 spectral terms (e.g., *Bright*, *Dull/Dark*, *Muddy*, *Boomy*, *Thin*, *Boxy*, *Nasal*, *Harsh*, *Sibilant*, *Airy*), 4 spatial terms (e.g., *Spacious*, *Distant*, *Dry*, *Wet*), and 4 dynamic terms (e.g., *Punchy*, *Compressed*, *Flat*, *Upfront/Forward*).

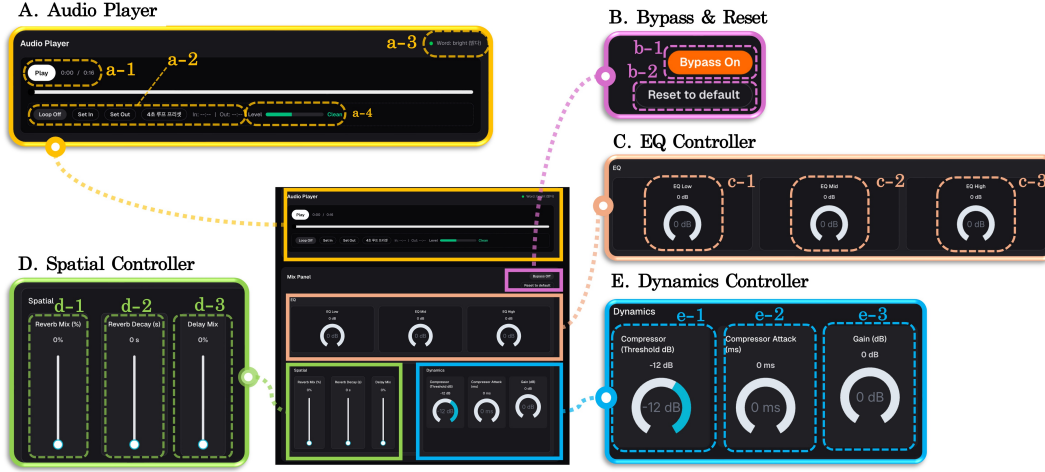


Figure 2: The web-based mixing interface used in the experiment. (A) Audio Player & Task Display, (B) Utility Tools (Bypass/Reset), (C) EQ Controller, (D) Spatial Controller (Reverb/Delay), and (E) Dynamics Controller.

2.4 Procedure

The experiment was conducted in participants' personal studios, allowing them to leverage familiar monitoring chains and minimize environmental bias. Participants adjusted 9 parameters for randomized descriptors via the web interface to match the target characteristics. Researcher intervention was strictly limited to technical support to prevent observer bias.

2.5 Data Analysis

2.5.1 3D Semantic Space Construction. To simplify the analysis of complex mix parameters, we employed Principal Component Analysis (PCA) to reduce nine technical variables into a perceivable 3D coordinate system. PCA identifies the primary dimensions of variation, mapping high-dimensional technical data onto three intuitive perceptual axes: *Spatial* (PC1), *Dynamics* (PC2), and *Spectral* (PC3). Within this space, the eight expert vectors for each descriptor form a geometric cluster. We quantified the degree of expert consensus by calculating the Convex Hull volume of these clusters, where a smaller volume indicates higher consensus and a larger volume suggests interpretative divergence.

2.5.2 Classification of Ambiguity Types. To categorize the patterns of ambiguity, we analyzed the **displacement vectors** relative to a neutral reference state (e.g., flat EQ, 0dB gain). For each descriptor, we aggregated the set of 8 displacement vectors into two dimensions: (1) *Directional Dispersion* (x -axis), calculated as $1 - \text{Mean Cosine Similarity}$ among the displacement vectors. This quantifies disagreement in the *direction* of adjustment (e.g., boosting lows vs. cutting highs) regardless of intensity; (2) *Intensity Variance* (y -axis), calculated as the variance of displacement magnitudes (L2 norm) from the neutral origin. This represents a disagreement on the *amount* of adjustment applied. To ensure an objective categorization, we classified these patterns using non-supervised **Gaussian Mixture Model (GMM)** clustering ($k = 3$) [4], accommodating

inconsistent variances, grouping descriptors that share similar dispersion profiles rather than imposing rigid geometric boundaries.

2.5.3 Ambiguity Decomposition. To quantify whether ambiguity is concentrated within a specific parameter or distributed across all dimensions, we calculated the Ambiguity Concentration using the **Gini coefficient**, a statistical measure of inequality that quantifies dispersion among values [17]. While traditionally an economic metric for inequality, we adapted it to gauge the **concentration of variance** by treating the three semantic axes (Spatial, Dynamics, Spectral) as the population and their respective variance contributions as wealth [14, 41]. In this context, a high Gini coefficient indicates **localized ambiguity** driven by conflicting strategies (e.g., Spectral vs. Spatial), whereas a low coefficient suggests **global ambiguity** where uncertainty is pervasive across all domains.

3 Results

We analyzed the distribution of expert responses for each descriptor, utilizing Min-Max normalization ($[0, 1]$) to unify heterogeneous parameter scales. PCA confirmed that the semantic space aligns with standard mixing dimensions, Spectral, Spatial, and Dynamics (see Appendix A for details). As illustrated in Fig. 3, descriptors such as *Punchy* ($V = 0.0008$), *Bright* ($V = 0.0024$), and *Dry* ($V = 0.0035$) formed relatively tight clusters, while terms like *Dull/Dark* ($V = 0.0224$), *Spacious* ($V = 0.0285$), and *Boomy* ($V = 0.0286$) occupied significantly larger volumes, reflecting divergence in expert interpretations. For the complete visualization of all 18 descriptors, see Appendix A. We clustered the descriptors based on their directional dispersion and magnitude variance, as shown in Fig. 4(a). The descriptors were categorized into three distinct types of ambiguity. **Type 1 (High Consensus)** was characterized by low dispersion in both magnitude and direction, indicating a singular, agreed-upon definition among experts. **Type 2 (Intensity Variance)** showed relatively low directional dispersion but high magnitude variance. Experts agree on *what* parameters to adjust

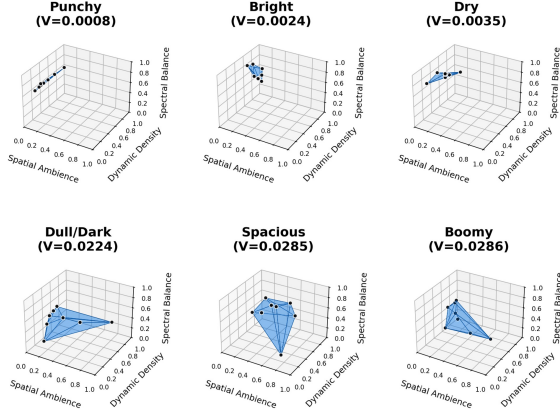


Figure 3: 3D Semantic Space of Mixing Descriptors. Each point represents an individual expert’s parameter setting projected into the semantic space.

but differed on *how much* intensity is required. **Type 3 (Directional Divergence)** exhibited high directional dispersion. This implies fundamental disagreements in the approach itself, representing the highest level of ambiguity where experts diverge in their problem-solving strategies.

The decomposition of ambiguity via the Gini coefficient, illustrated in Fig. 4(b), reveals that the *structure* of disagreement offers deeper insights than the total variance magnitude alone. Overall terms in Type 3 had relatively high total variance, but there were exceptions, showing that high magnitude does not strictly correlate with high ambiguity. For instance, *Wet* (Type 1) exhibited large total variance (0.09) due to the high sensitivity of spatial parameters, despite expert consensus on the direction. Conversely, *Boxy* (Type 3) showed low total variance (0.036), indicating that experts diverged in strategy but applied only subtle adjustments to avoid signal degradation. Crucially, within Type 3 (Directional Divergence), the Gini coefficient uncovers two distinct structural behaviors. *Boxy* recorded the lowest Gini coefficient ($G = 0.10$), indicating that the variance is evenly distributed across the Spatial (39%), Spectral (38%), Dynamics (23%) domains. This suggests **global ambiguity**, where experts lack a dominant strategy and resort to diverse, multi-domain approaches. In contrast, *Boomy* exhibited a notably higher Gini coefficient ($G = 0.36$), implying that disagreement is not random but **concentrated within specific axes**—primarily driven by a conflict between Spectral and Spatial processing. This empirically supports prior findings characterizing *Muddy* as a complex descriptor [10, 35], while further clarifying that its core conflict arises specifically from the trade-off between competing mixing strategies.

4 Discussions

4.1 Rethinking Semantic Mapping

This study moves beyond the premise that specific words hold a single, fixed interpretation, confirming that multiple interpretations for the same descriptor can coexist depending on the engineer’s experience and context. Our findings on *High Consensus* terms align

with prior large-scale studies [6, 35, 42], confirming that terms like *Bright* or *Wet* target specific parameters, whereas our analysis of divergence offers new insights. Crucially, complex terms transcended category boundaries; for instance, *Airy* (Spectral) required combinatorial adjustments in both High EQ and Reverb, differing from the suppression of reverb in *Bright*. This suggests that divergence is not statistical noise but an intrinsic characteristic where perceptual goals require multi-dimensional strategies. Therefore, interfaces must move beyond deterministic mappings to accommodate this structured ambiguity.

4.2 Design Implications

We envision a collaborative communication support system designed to align novice vocabulary and mental models with expert standards [1, 8, 18, 24] by visualizing intent prior to transmission [39]. Prior to the iterative feedback process with an engineer, a novice can utilize this system to refine their auditory goals. Addressing the limitations of static text-to-parameter mappings [11, 38], our adaptive framework manages ambiguity [15, 16] by dynamically balancing system efficiency and user agency based on the analyzed types [20, 33].

Automation for High Consensus. For terms with standardized definitions (e.g., *Bright*), manual adjustment imposes unnecessary load [32]. Leveraging *functional abstraction* [28], the system should offer **one-shot presets**. This serves as a reliable building block [27], allowing novices to engage in rapid prototyping [2].

Degree Control for Intensity Variance. Where experts agree on direction but differ on magnitude [40], we apply *constraints* [30] by compressing parameters into a **single macro slider**. This one-to-many mapping replaces high-dimensional complexity with perceptual control, preserving user agency over intensity [19].

Exploratory Interface for Directional Divergence. For ambiguous terms where strategies conflict (e.g., Option A vs. Option B), deterministic mappings risk failure [34]. We propose an **exploratory strategy** [7, 25] presenting diverse candidates generated from data clusters. This leverages recognition over recall [9, 29, 30] to aid sensemaking and concretize intent [15, 33].

By interacting with these dynamically allocated widgets, novices can effectively translate their auditory goals into precise technical instructions. To illustrate, inputting a high-consensus term like *Bright* instantly applies a predefined EQ preset (Type 1). If they want the track more *Distant*, they adjust a 0-100% macro slider rather than grappling with complex reverb ratios (Type 2). Finally, for a divergent term like *Muddy*, the system presents an A/B gallery—such as cutting low-mids versus altering compression—enabling the user to compare previews and select the exact approach that matches their intent (Type 3).

5 Conclusions

This study unpacks the black box of mixing ambiguity, demonstrating that expert disagreement is not random noise but a structured diversity of perspectives. By categorizing ambiguity into three types, we proposed a design framework that adapts its interface to the nature of the user’s intent. While our current study is limited by a relatively small sample size and homogeneous acoustic stimuli, we expect future experiments with larger, genre-diverse datasets to

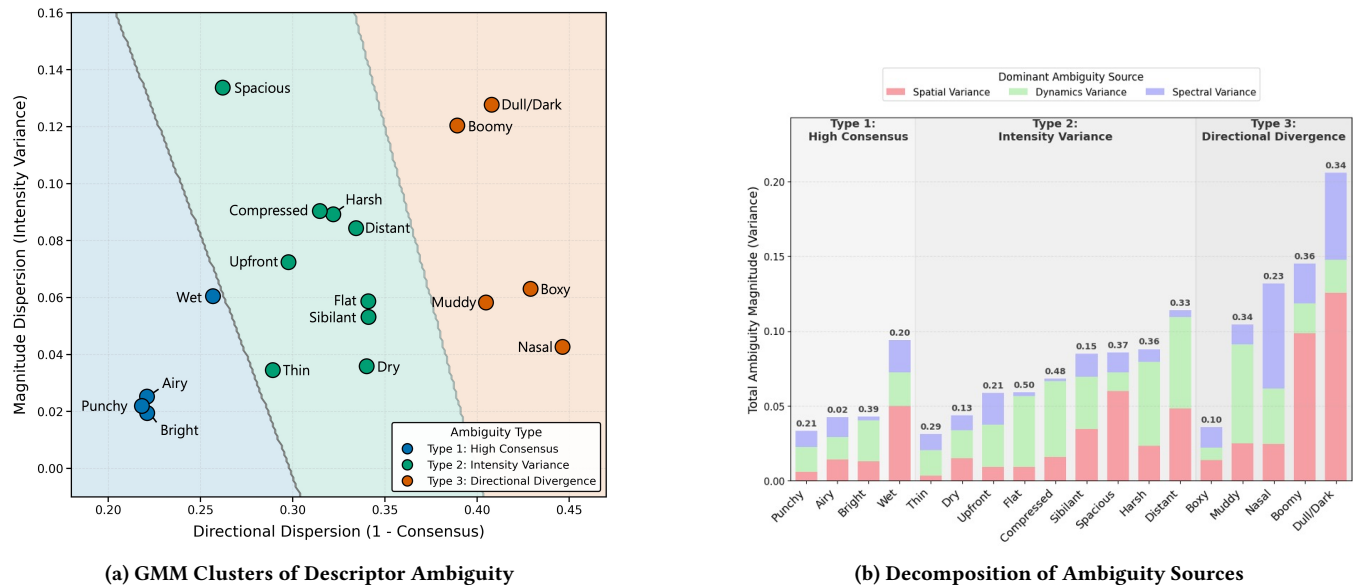


Figure 4: Structural Analysis of Semantic Ambiguity. (a) Expert disagreement is classified into three distinct patterns: High Consensus, Intensity Variance, and Directional Divergence. (b) Decomposing variance reveals the source of ambiguity

establish a broader range of adaptive interface strategies. Building on the framework suggested in this study, we plan to expand our research to a comparative user study involving novice musicians to empirically validate whether this adaptive approach effectively reduces the rounds of revision and enhances creative agency in remote collaboration scenarios.

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF) grants funded by the Korea government (MSIT) (RS-2024-00343397; RS-2025-16902996).

References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [2] Nattaphob Aoonlamai and Parama Kwangmuang. 2025. Integrating digital tools and constructivist learning: a ubiquitous learning framework for enhancing creativity in music education. *BMC Psychology* 13, 1 (Sept. 2025), 18 pages. doi:10.1186/s40359-025-03300-z
- [3] Adam Bell, Ethan Hein, and Jarrod Ratcliffe. 2015. Beyond skeuomorphism: The evolution of music production software user interface metaphors. *Journal on the Art of Record Production* 9 (01 2015).
- [4] Christopher M. Bishop. 2006. *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Springer-Verlag, Berlin, Heidelberg.
- [5] R.J. Burgess. 2013. *The Art of Music Production: The Theory and Practice*. OUP USA, New York. https://books.google.co.kr/books?id=m4dNEAAQBAJ[https://books.google.co.kr/books?id=m4dNEAAQBAJ]
- [6] Mark Cartwright and Bryan Pardo. 2013. Social-EQ: Crowdsourcing an Equalization Descriptor Map. In *Proceedings of the 14th International Society for Music Information Retrieval Conference (ISMIR)*. ISMIR, Curitiba, Brazil, 395–400. https://brianmcfree.net/ismir2013/wp-content/uploads/2013/09/106_Paper.pdf
- [7] Mark Cartwright, Bryan Pardo, and Josh Reiss. 2014. MIXPLORATION: rethinking the audio mixer interface. In *Proceedings of the 19th International Conference on Intelligent User Interfaces* (Haifa, Israel) (IUI '14). Association for Computing Machinery, New York, NY, USA, 365–370. doi:10.1145/2557500.2557530
- [8] Herbert H. Clark and Susan E. Brennan. 1991. *Grounding in communication*. American Psychological Association, Washington, DC, 127–149. doi:10.1037/10096-006
- [9] Michael A. Cohen, Karla K. Evans, Todd S. Horowitz, and Jeremy M. Wolfe. 2011. Auditory and visual memory in musicians and nonmusicians. *Psychonomic Bulletin & Review* 18, 3 (03 2011), 586–591. doi:10.3758/s13423-011-0074-0
- [10] Brecht De Man, Matthew Boerum, Brett Leonard, Richard King, George Massenburg, and Joshua D. Reiss. 2015. Perceptual Evaluation of Music Mixing Practices. In *Proceedings of the 138th Audio Engineering Society Convention (AES)*. Audio Engineering Society, Warsaw, Poland, 9235. https://api.semanticscholar.org/CorpusID:109769061
- [11] Brecht De Man and Joshua Reiss. 2013. A Semantic Approach To Autonomous Mixing. *Journal of the Art of Record Production* 8 (12 2013).
- [12] Brecht De Man and Joshua D. Reiss. 2017. The Mix Evaluation Dataset. In *Proceedings of the 20th International Conference on Digital Audio Effects (DAFx-17)*. University of Edinburgh, Edinburgh, UK, 436–442.
- [13] Felix Duvallet, Matthew Walter, Thomas Howard, Sachithra Hemachandra, Jean Oh, Seth Teller, Nicholas Roy, and Anthony Stentz. 2016. *Inferring Maps and Behaviors from Natural Language Instructions*. Vol. 109. Springer Nature, Cham, 373–388. doi:10.1007/978-3-319-23778-7_25
- [14] Ben Fauber. 2024. Gini Coefficient as a Unified Metric for Evaluating Many-versus-Many Similarity in Vector Spaces. doi:10.48550/ARXIV.2411.07983
- [15] Tong Gao, Mira Dontcheva, Eytan Adar, Zhicheng Liu, and Karrie G. Karahalios. 2015. DataTone: Managing Ambiguity in Natural Language Interfaces for Data Visualization. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology* (Charlotte, NC, USA) (UIST '15). Association for Computing Machinery, New York, NY, USA, 489–500. doi:10.1145/2807442.2807478
- [16] William W. Gaver, Jacob Beaver, and Steve Benford. 2003. Ambiguity as a resource for design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Ft. Lauderdale, Florida, USA) (CHI '03). Association for Computing Machinery, New York, NY, USA, 233–240. doi:10.1145/642611.642653
- [17] Joe Hasell. 2023. Measuring inequality: what is the Gini coefficient? Our World in Data. https://ourworldindata.org/what-is-the-gini-coefficient
- [18] Cindy E. Hmelo-Silver and Merav Green Pfeiffer. 2004. Comparing expert and novice understanding of a complex system from the perspective of structures, behaviors, and functions. *Cognitive Science* 28, 1 (2004), 127–138. doi:10.1016/S0364-0213(03)00065-X
- [19] Andy Hunt, Marcelo M. Wanderley, and Ross Kirk. 2000. Towards a Model for Instrumental Mapping in Expert Musical Interaction. In *Proceedings of the 2000 International Computer Music Conference (ICMC 2000)*. International Computer Music Association, Berlin, Germany, 4 pages.
- [20] Edwin L. Hutchins, James D. Hollan, and Donald A. Norman. 1985. Direct Manipulation Interfaces. *Human-Computer Interaction* 1, 4 (1985), 311–338. arXiv:https://doi.org/10.1207/s15327051hci0104_2 doi:10.1207/s15327051hci0104_2

- [21] ITU-R. 2015. *Recommendation BS.1770-4: Algorithms to measure audio programme loudness and true-peak audio level*. Recommendation. International Telecommunication Union, Geneva.
- [22] Roey Izhaki. 2017. *Mixing Audio: Concepts, Practices, and Tools* (3rd ed.). Routledge, New York, NY, USA. doi:10.4324/9781315666668
- [23] Nicholas Jillings, David Moffat, Brecht De Man, and Joshua D. Reiss. 2015. Web Audio Evaluation Tool: A Browser-Based Listening Test Environment. In *Proceedings of the 12th Sound and Music Computing Conference (SMC 2015)*. Sound and Music Computing Network, Maynooth, Ireland, 143–148.
- [24] Kangbeen Ko, Minwoo Oh, Minwoo Seong, and SeungJun Kim. 2025. LEGO-LAS: Learning & Enhancing Golf Skills through LLM-Augmented System. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 368, 10 pages. doi:10.1145/3706599.3720141
- [25] Brian Lee, Savil Srivastava, Ranjitha Kumar, Ronen Brafman, and Scott R. Klemmer. 2010. Designing with interactive example galleries. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI '10). Association for Computing Machinery, New York, NY, USA, 2257–2266. doi:10.1145/1753326.1753667
- [26] Justin Matejka, Michael Glueck, Erin Bradner, Ali Hashemi, Tovi Grossman, and George Fitzmaurice. 2018. Dream Lens: Exploration and Visualization of Large-Scale Generative Design Datasets. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3173574.3173943
- [27] Evandro Manara Miletto, Marcelo Soares Pimenta, François Bouchet, Jean-Paul Sansonnet, and Damián Keller. 2011. Principles for Music Creation by Novices in Networked Music Environments. *Journal of New Music Research* 40, 3 (Sept. 2011), 205–216. doi:10.1080/09298215.2011.603832
- [28] Marie Muehlhaus, Alexander Liggesmeyer, and Jürgen Steimle. 2025. ExoKit: A Toolkit for Rapid Prototyping of Interactions for Arm-based Exoskeletons. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 282, 17 pages. doi:10.1145/3706598.3713815
- [29] Jakob Nielsen. 1995. Usability inspection methods. In *Conference Companion on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '95). Association for Computing Machinery, New York, NY, USA, 377–378. doi:10.1145/223355.223730
- [30] Donald A. Norman. 2002. *The Design of Everyday Things*. Basic Books, Inc., USA.
- [31] Bobby Owsinski. 2017. *The Mixing Engineer's Handbook* (4th ed.). Bobby Owsinski Media Group, Burbank, CA, USA.
- [32] Mitchel Resnick, Brad Myers, Kumiyo Nakakoji, Ben Shneiderman, Randy Pausch, and Mike Eisenberg. 2005. Design Principles for Tools to Support Creative Thinking. *Report of Workshop on Creativity Support Tools* 20 (01 2005).
- [33] Ben Shneiderman. 2007. Creativity support tools: accelerating discovery and innovation. *Commun. ACM* 50, 12 (Dec. 2007), 20–32. doi:10.1145/1323688.1323689
- [34] Ryan Stables, Brecht De Man, Sean Enderby, Joshua D. Reiss, György Fazekas, and Thomas Wilmering. 2016. Semantic Description of Timbral Transformations in Music Production. In *Proceedings of the 24th ACM International Conference on Multimedia* (Amsterdam, The Netherlands) (MM '16). Association for Computing Machinery, New York, NY, USA, 337–341. doi:10.1145/2964284.2967238
- [35] Ryan Stables, Sean Enderby, Brecht De Man, György Fazekas, and Joshua Reiss. 2014. SAFE: A System for the Extraction and Retrieval of Semantic Audio Descriptors. In *Proceedings of the 15th International Society for Music Information Retrieval Conference (ISMIR 2014)*. International Society for Music Information Retrieval, Taipei, Taiwan, 2 pages.
- [36] Matúš Sulir and Jaroslav Porubán. 2021. Natural mapping between voice commands and APIs. *Open Computer Science* 11, 1 (Jan. 2021), 135–145. doi:10.1515/comp-2020-0125
- [37] Soumya Sai Vanka, Maryam Safi, Jean-Baptiste Rolland, and György Fazekas. 2023. The Role of Communication and Reference Songs in the Mixing Process: Insights from Professional Mix Engineers. arXiv:2309.03404 [cs.HC] https://arxiv.org/abs/2309.03404[https://arxiv.org/abs/2309.03404]
- [38] Satvik Venkatesh, David Moffat, and Eduardo Reck Miranda. 2022. Word Embeddings for Automatic Equalization in Audio Mixing. *Journal of the Audio Engineering Society* 70, 9 (Nov. 2022), 753–763. doi:10.17743/jaes.2022.0047
- [39] Jo Vermeulen, Kris Luyten, Elise van den Hoven, and Karin Coninx. 2013. Crossing the bridge over Norman's Gulf of Execution: revealing feedforward's true identity. In *CHI '13: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 1931–1940. doi:10.1145/2470654.2466255
- [40] Asteris Zacharakis, Konstantinos Pasiadis, and Joshua Reiss. 2015. An Inter-language Unification of Musical Timbre: Bridging Semantic, Perceptual, and Acoustic Dimensions. *Music Perception: An Interdisciplinary Journal* 32 (04 2015), 394–412. doi:10.1525/mp.2015.32.4.394
- [41] Yuchao Zhang, Bin Nie, Jianqiang Du, Jiandong Chen, Yuwen Du, Haik Jin, Xuepeng Zheng, Xingxin Chen, and Zhen Miao. 2023. Feature selection based on neighborhood rough sets and Gini index. *PeerJ Computer Science* 9 (Dec. 2023), e1711. doi:10.7717/peerj-cs.1711
- [42] Taylor Zheng, Prem Seetharaman, and Bryan Pardo. 2016. SocialFX: Studying a Crowdsourced Folksonomy of Audio Effects Terms. In *Proceedings of the 24th ACM international conference on Multimedia* (Amsterdam, The Netherlands) (MM '16). Association for Computing Machinery, New York, NY, USA, 182–186. doi:10.1145/2964284.2967207